KEMENTERIAN PENDIDIKAN

KOLEJ VOKASIONAL
JABATAN TEKNOLOGI MAKLUMAT
PROGRAM DIPLOMA TEKNOLOGI KOMPUTERAN

PERANCANGAN KURSUS
COURSE OUTLINE

MAKLUMAT KURSUS (*COURSE INFORMATION*)

SEMESTER/TAHUN (*SEMESTER/YEAR*)	**: SEMESTER 1 TAHUN 2**
KOD KURSUS (*COURSE CODE*)	**: DKA3223**
NAMA KURSUS (*NAME OF COURSE*)	**: ARTIFICIAL INTELLIGENCE FOR COMPUTER VISION**

PEMBAHAGIAN JAM BELAJAR PELAJAR *(Distribution of Student Learning Time-SLT):*

Kategori Aktiviti *(Category of Activities)*	**Aktiviti Pembelajaran** *(Learning Activity)*		**Jumlah Jam/Semester** *(Total Hours/Semester)*
Pembelajaran Dan Pengajaran *(Learning and Teaching)*	Pembelajaran Berbantu *(Guided Learning (F2F))*	Kuliah *(Lecture)*	13.0
		Tutorial / Amali *(Tutorial / Practical)*	52.0
	Pembelajaran Kendiri *(Independent Learning (NF2F))*		28.0
		Jumlah Jam *(Sub total)*	**93.0**
Penilaian Berterusan *(Continuous Assessment)*	Bersemuka (F2F) *(Physical (F2F))*		5.0
	Kendiri (NF2F) *(Independent Learning (NF2F))*		11.2
		Jumlah Jam *(Sub total)*	**16.2**
Penilaian Akhir *(Final Assessment)*	Teori *(Theory)*	Bersemuka (F2F) *(Physical (F2F))*	-
		Kendiri (NF2F) *(Independent Learning (NF2F))*	-
	Amali *(Practical)*	Bersemuka (F2F) *(Physical (F2F))*	2.0
		Kendiri (NF2F) *(Independent Learning (NF2F))*	8.8
		Jumlah Jam *(Sub total)*	**10.8**
		JUMLAH JAM BELAJAR *(TOTAL STUDENT LEARNING TIME -SLT)*	**120**
		NILAI KREDIT *(CREDIT VALUE)*	**3**

Kursus Pra-syarat (*Pre-requisite Course*)	: None
Nama Pensyarah (*Name of Lecturer*)	:

Disediakan oleh (*Prepared by*)	: **PENYELARAS KURSUS**	**Disahkan oleh** (*Approved by*)	: **KETUA PROGRAM/ KETUA JABATAN**
Tandatangan (*Signature*)	:	**Tandatangan** (*Signature*)	:
Nama (*Name*)	: Ts Saraswathy Chandiran	**Nama** **(*Name*):**	:
Tarikh (*Date*):	:	**Tarikh** (*Date*)	:

MATLAMAT (*GOALS*):

This course is required to enhance students understanding in AI and edge computing concepts, explore the architecture and capabilities of the NVIDIA Jetson platform, and develop AI models optimized for edge deployment. They will learn to apply model optimization techniques, such as TensorRT acceleration, and work with key NVIDIA tools like DeepStream to enhance performance. Additionally, students will gain hands-on experience in deploying AI applications, troubleshooting challenges, and optimizing real-world implementations using NVIDIA's frameworks and best practices.

SINOPSIS (*SYNOPSIS*):

The course provides comprehensive training on NVIDIA's Jetson platform, designed for deploying AI at the edge. This course covers everything from an introduction to AI and edge computing concepts to advanced topics, including optimization and deployment of AI models using NVIDIA's tools and frameworks.

HASIL PEMBELAJARAN (*LEARNING OUTCOMES*):

Upon completion of the course, students will be able to:

1. Construct an understanding of the fundamentals of AI and edge computing, including the importance and application of AI at the edge. (C3, PLO3, C2)
2. Build AI models deployment on Jetson devices for real-time inference, applying practical skills in computer vision, object detection, and other edge computing applications. (P3, PLO2, C3a)
3. Initiate the Jetson development environment configuration and utilize PyTorch for model development, training, and optimization. (A3, PLO10, C3d)

ISI KANDUNGAN (*CONTENT*):

MINGGU (*WEEK*)	**KANDUNGAN** (*CONTENT*)	**PENTAKSIRAN** (*ASSESSMENT*)
1	**1.0 GIT AND GITHUB** 1.1 Introduction to version control systems 1.2 Key concepts: repositories, commits, branches, pull requests, and merging Computer network layout/diagram contents 1.3 GitHub platform overview and its role in collaboration	• Written Test • Practical Test 1
2	**2.0 PYTHON PROGRAMMING** 2.1 Python basics: syntax, variables, data types, and control structures 2.2 Libraries for AI: NumPy, pandas, and matplotlib 2.3 Writing modular and reusable code	• Written Test • Practical Test 1
3	**3.0 AI CODE ASSISTANT (E.G., CURSOR AI, CHATGPT)** 3.2 Overview of AI code assistants and their applications 3.3 Benefits and limitations in software development	• Written Test • Practical Test 1
4	**4.0 ARTIFICIAL NEURAL NETWORKS (ANN) [CONCEPT]** 4.1 Introduction to ANN: structure and components (neurons, layers, weights, and activation functions) 4.2 Forward and backward propagation 4.3 Common applications of ANN Recording	• Written Test

5	**5.0 ARTIFICIAL NEURAL NETWORKS (ANN)** 5.1 Development of ANN using PyTorch 5.2 Hyperparameter tuning and modelling	• Practical Test 1 • Practical Test 2
6	**6.0 CONVOLUTIONAL NEURAL NETWORKS (CNN) [CONCEPT]** 6.1 Understanding CNN architecture: convolutional layers, pooling, and fully connected layer 6.2 Applications in image processing	• Written Test
7	**7.0 CONVOLUTIONAL NEURAL NETWORKS (CNN) [PRACTICAL LAB]** 7.1 Development of CNN using PyTorch 7.2 Hyperparameter tuning and modelling	• Practical Test 1 • Practical Test 2
8	**8.0 OBJECT DETECTION [CONCEPT]** 8.1 Introduction to object detection and its real-world applications 8.2 Overview of object detection models (e.g., YOLO, Faster R-CNN)	• Written Test
9	**9.0 OBJECT DETECTION [PRACTICAL LAB - YOLO] (5 Hours)** 9.1 Development of YOLO using PyTorch 9.2 Hyperparameter tuning and modelling	• Written Test • Practical Test 2
10	**10.0 MODEL DEPLOYMENT (FASTAPI, PYDANTIC)** 10.1 Introduction to FastAPI and its advantages for API development 10.2 Overview of Pydantic for data validation and serialization 10.3 Using FastAPI to deploy AI models for real-time inference	• Written Test
11	**11.0 NVIDIA LIBRARY (DEEPSTREAM, TAO TOOLKIT, TRITON)** 11.1 Overview of NVIDIA libraries and their use in AI workflows 11.2 Features of DeepStream, TAO Toolkit, and Triton	• Written Test
12	**12.0 AIOT [PRACTICAL LAB]** 12.1 Introduction to AI and IoT integration 12.2 Real-world applications of AIoT in automation and monitoring	

13	**13.0 GROUP PROJECT** 13.1 Overview of project requirements and deliverables 13.2 Recap of key concepts learned throughout the course	
14	**14.0 INDIVIDUAL PROJECT AND COURSE WRAP-UP** 14.1 Review and finalize Individual projects 14.2 Reflections on the course and discussion on future learning paths	
15	**PREPARATION WEEK**	
16	**PRACTICAL / FINAL EXAMINATION**	
17	**PRACTICAL / FINAL EXAMINATION**	

AMALI (*PRACTICAL*):

1. Perform a basic deep learning model using structured data.
2. Perform a simple object detection project

PENILAIAN (*ASSESSMENT*):

<table>
<tr><th rowspan="2">No</th><th rowspan="2">Type of Assessment</th><th colspan="3">CLO</th><th rowspan="2">Sub Total</th><th rowspan="2">Total</th></tr>
<tr><th>CLO 1:
C3, PLO3, C2
(Cognitive)</th><th>CLO 2:
P4, PLO2, C3a
(Psychomotor)</th><th>CLO 3:
A2, PLO7, C3d
(Affective)</th></tr>
<tr><td rowspan="4">1</td><td colspan="5">Continuous Assessment</td><td rowspan="4">60%</td></tr>
<tr><td>Written Test</td><td>20%</td><td>-</td><td>-</td><td>10%</td></tr>
<tr><td>Practical Test 1</td><td>-</td><td>20%</td><td>-</td><td rowspan="2">40%</td></tr>
<tr><td>Practical Test 2</td><td>-</td><td>20%</td><td>-</td></tr>
<tr><td rowspan="2">2</td><td colspan="5">Final Assessment</td><td rowspan="2">40%</td></tr>
<tr><td>Project/ Assignment</td><td>-</td><td>20%</td><td>20%</td><td>40%</td></tr>
<tr><td colspan="2">Total Mark (CLO)</td><td>20%</td><td>60%</td><td>20%</td><td colspan="2">100%</td></tr>
</table>

Assessment Specifications Table (AST):

CONTEXT	CLO	ASSESSMENT METHODS FOR COURSEWORK (CA)

	CLO1	CLO2	CLO3	WRITTEN TEST	PRACTICAL TEST 1	PRACTICAL TEST 2
1.0 Git and GitHub	√	√	√			–
2.0 Python Programming	√	√	√	√		-
3.0 AI Code Assistant (e.g., Cursor AI, ChatGPT)	√	√	√		√	-
4.0 Artificial Neural Networks (ANN) [Concept]	√	–	√			-
5.0 Artificial Neural Networks (ANN) [Practical Lab]	–	√	√	-		√
6.0 Convolutional Neural Networks (CNN) [Concept]	√	–	√	√		–
7.0 Convolutional Neural Networks (CNN) [Practical Lab]	–	√	√	-	√	√
8.0 Object Detection [Concept]	√	–	√	√	-	-
9.0 Object Detection [Practical Lab - YOLO]	–	√	√	-	-	√
10.0 Model Deployment (FastAPI, Pydantic)	√	–	√		-	-
11.0 NVIDIA Library (DeepStream, TAO Toolkit, Triton)	√	–	√	√	-	-
12.0 AIoT [Practical Lab]	–	–	√	-	-	-
13.0 Project	–	–	√	-	-	-
14.0 Project and Course Wrap-Up	–	–	√	-	-	-

Remark:

1. ***Suggested time for***
   - ***Written Test : 60 minutes***
   - ***Practical Test 1 : 120 minutes***
   - ***Practical Test 2 : 120 minutes***
   - ***Project / Assignment : 120* minutes**
2. ***40 Notional hours is equivalent to 1 credit.***

RUJUKAN (*REFERENCES*):

MAIN REFERENCE:

1. Caleb Kaplan (2024). Mastering Wireshark: Unveiling the Secrets of Network Analysis, USA: Independently published, ISBN-13: 979-8877245570
2. Joseph Migga Kizza (2020). Guide to Computer Network Security (Texts in Computer Science) 5th Edition, USA: Springer International Publishing AG.
3. Aditya Mukherjee (2020). Network Security Strategies: Protect your network and enterprise against advanced cybersecurity attacks and threats 1st Edition, Birmingham: Packt Publishing Ltd

KEHADIRAN/PERATURAN SEMASA KULIAH (*LECTURE ATTENDANCE/REGULATION*):

1. Pelajar mesti hadir tidak kurang dari 80% masa pertemuan yang ditentukan bagi sesuatu kursus.
   Students must attend lectures not less than 80% of the contact hours for every course.

2. Pelajar yang tidak memenuhi perkara (1) di atas tidak dibenarkan menghadiri kuliah dan menduduki sebarang bentuk penilaian selanjutnya. Markah sifar (0) akan diberikan kepada pelajar yang gagal memenuhi perkara (1).
   Students who do not fulfill (1) will not be allowed to attend further lectures and sit for any further

examination. Zero mark (0) will be given to students who fail to comply with (1).

3. Pelajar perlu mengikut dan patuh kepada peraturan berpakaian yang berkuatkuasa dan menjaga disiplin diri masing-masing untuk mengelakkan dari tindakan tatatertib diambil terhadap pelajar.
   Students must obey all rules and regulations of the university and must discipline themselves in order to avoid any disciplinary actions against them.

4. Pelajar perlu mematuhi peraturan keselamatan semasa proses pembelajaran dan pengajaran.
   Student must obey safety regulations during learning and teaching process.

MATRIK HASIL PEMBELAJARAN KURSUS DAN HASIL PEMBELAJARAN PROGRAM
(*MATRIX OF COURSE LEARNING OUTCOMES AND PROGRAMME LEARNING OUTCOMES*)

Dilampirkan (*As attached*).

MATRIK HASIL PEMBELAJARAN KURSUS DAN HASIL PEMBELAJARAN PROGRAM
MATRIX OF COURSE LEARNING OUTCOMES AND PROGRAMME LEARNING OUTCOMES

Nama PPT: Kolej Vokasional………………., Kementerian Pendidikan Malaysia
Program: Diploma Teknologi Komputeran
Kursus: Artificial Intelligence For Computer Vision
Kod: DKA3223

Matrik ini perlu dilampirkan bersama
1. Objektif Pendidikan Program (PEO)
2. Hasil Pembelajaran Program (PLO)

Bil. (*No.*)	**Hasil Pembelajaran Kursus** (*Course Learning Outcomes*)	**Pematuhan kepada PLO** (*Compliance to PLO*)											**Kaedah Penyampaian** (*Method of Delivery*)	**Kaedah Pentaksiran** (*Method of Assessment*)	**KPI**
		PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**			
1	Construct an understanding of the fundamentals of AI and edge computing, including the importance and application of AI at the edge. (C3, PLO3, C2)	C3											Case Study, Project, Tutorial, Group Work, SCL, Lab	Written Test	100% of students scored more than 50%
2	Build AI models deployment on Jetson devices for real-time inference, applying practical skills in computer vision, object detection, and other edge computing applications. (P3, PLO2, C3a)		P3										Practical, Demonstration	Practical Test 1, Practical Test 2, Final Assessment (Project)	100% of students scored more than 50%
3	Initiate the Jetson development environment configuration and utilize PyTorch for model development, training, and optimization. (A3, PLO10, C3d)						A3						Case Study, Project	Final Assessment (Project)	100% of students scored more than 50%
	Total	**1**	**1**				**1**								

PEMETAAN PLO – MQF2.0 LOD – MBOT LOD			LEVEL OF LEARNING TAXONOMY		
PROGRAMME LEARNING OUTCOME (PLO)	**MQF2.0 LEARNING OUTCOME DOMAIN (LOD)**	**MBOT LEARNING OUTCOME DOMAIN**	**COGNITIVE**	**PSYCHOMOTOR**	**AFFECTIVE**
PLO1 Pengetahuan & Kefahaman	**C1** Knowledge & Understanding	**MBOT1** Knowledge (Cognitive Domain)	**C1** Remember **C2** Understand **C3** Apply **C4** Analyze **C5** Evaluate **C6** Create	**P1** Perception **P2** Set **P3** Guided Response **P4** Mechanism **P5** Complex Overt Response **P6** Adaptation **P7** Origination	**A1** Receiving **A2** Responding **A3** Valuing **A4** Organising **A5** Internalising
PLO2 Kemahiran praktikal	**C3a** Practical Work Skills	**MBOT2** Practical Skills/Modern Tool Usage/DigitalSkills (Psychomotor Domain)			
PLO3 Kemahiran Saintifik & Penyelesaian masalah	**C2** Cognitive Skill	**MBOT3** Analytical, Critical Thinking, Design Thinking and Scientific Approach/Numeracy Skills (Cognitive Domain)			
PLO4 Kemahiran berkomunikasi	**C3c** Communication Skills	**MBOT4** Communication Skills (Affective Domain)			
PLO5 Kemahiran Sosial & Kerja Berpasukan	**C3b** Interpersonal Skills	**MBOT5** Social And Responsibility In Society And Technologist Community (Affective Domain)			
PLO6 Kemahiran Personal	**C4a** Personal Skills	**MBOT6** Lifelong Learning And Information Management/Personal Skills (Affective Domain)			
PLO7 Kemahiran Keusahawanan	**C4b** Entrepreneurship Skills	**MBOT7** Technoprenuership And Management Skills (Affective Domain)			
PLO8 Etika & Profesionalisme	**C5** Ethics and Professionalism	**MBOT8** Ethics And Professionalism (Affective Domain)			
PLO9 Kepimpinan, Autonomi & Tanggungjawab	**C3f** Leadership, Autonomy and Responsibility Skills	**MBOT9** Teamwork And Leadership (Affective Domain)			
PLO10 Kemahiran Digital	**C3d** Digital Skills	**MBOT2** Practical Skills/Modern Tool Usage/DigitalSkills (Psychomotor Domain)			
PLO11 Kemahiran Numerasi	**C3e** Numeracy Skills	**MBOT3** Analytical, Critical Thinking, Design Thinking and Scientific Approach/Numeracy Skills (Cognitive Domain)			